

An analytic proof for two colors

Common pair averages with an audited Gaussian input

Working research notes

18 September 2026

Abstract

We prove the two-color theorem using the older Frantzikinakis–Host Gaussian cancellation theorem and scalar pretentious concentration. The Gaussian reduction, including fixed affine grids, is given explicitly. The disputed Frantzikinakis–Mountakis four-point recurrence theorem and its wrapped-phase localization are not used. The argument uses an explicit common CRT and angular averaging scheme, and the ordinary two-point spectral theorem. It does not invoke the computer-assisted two-color theorem. The written derivation and Gaussian input reduction have passed independent internal audits. No claim of novelty is made; it does not prove the assertion for an arbitrary finite number of colors.

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1 A two-color proof through common pair averages

The source audit replaces the newer generalized cancellation citation by the older Frantzikinakis–Host Gaussian theorem, with the explicit affine-grid reduction supplied in Section 2. Together with the scalar concentration estimates accepted by the user, this proves the result below. The disputed four-point recurrence theorem and its wrapped-phase argument are not used. The proof uses the ordinary spectral theorem for an abelian group. It does not invoke the computer-assisted two-color theorem. No claim of novelty is made. All three pair correlations are controlled on *the same positive averages*.

Theorem 1.1. *Every $c : \mathbb{N} \rightarrow \{-1, 1\}$ admits positive integers x, y, z with $x^2 + y^2 = z^2$ and $c(x) = c(y) = c(z)$.*

Let $\mathcal{M} = (\widehat{\mathbb{Q}_{>0}}, \times)$, identified with the unit-modulus completely multiplicative functions on $\mathbb{N}$, extended by $f(a/b) = f(a)\overline{f(b)}$. This is a compact metrizable space under pointwise convergence. Set

$$\mathcal{E} = \{f \in \mathcal{M} : f(p) = \chi(p)p^{it} \text{ at all sufficiently large primes } p \text{ for some } \chi, t\},$$

where χ is a Dirichlet character and $t \in \mathbb{R}$. We construct common positive averages of

$$P_1(M, N) = 2MN, \quad P_2(M, N) = M^2 - N^2, \quad P_3(M, N) = M^2 + N^2, \quad M > N > 0,$$

whose scalar kernels $f(P_1/P_2), f(P_1/P_3), f(P_2/P_3)$ have the respective integrated limits

$$\mathbf{1}_{\{f=1\}}, \quad \mathbf{1}_{\{f=1\}}, \quad \mathbf{1}_{\mathcal{E}}(f). \quad (1)$$

Their nonnegativity will contradict the binary sign identity.

1.1 Precisely stated public analytic inputs

Put

$$\mathbb{D}(f, g; Y, \infty)^2 = \sum_{p > Y} \frac{1 - \Re(f(p)\overline{g(p)})}{p}.$$

Call f pretentious if $f \sim \chi n^{it}$, meaning $\mathbb{D}(f, \chi n^{it}; 1, \infty) < \infty$, for some χ, t ; call it aperiodic otherwise. This is the convention of the cancellation theorem used here. We may and do choose the Dirichlet models primitive, since replacing a character by its primitive associate changes only finitely many prime values. Thus its conductor is also its period.

Cancellation. Frantzikinakis–Host, [1, Theorem 9.7], with the fixed periodic-grid reduction in Section 2, gives

$$\mathbb{E}_{m, n \leq X} \mathbf{1}_S(Qm + a, Qn + b) \prod_{j=1}^s f_j(L_j(Qm + a, Qn + b)) g(P(Qm + a, Qn + b)) \longrightarrow 0. \quad (2)$$

The data Q, a, b are fixed, S is a homogeneous convex sector, $P(m, n) = m^2 + n^2$, and the unit-modulus completely multiplicative functions are extended evenly to negative arguments. An aperiodic f_1 occurs on a nonzero linear form L_1 independent of every other L_j . We use the forms $m, n, m - n, m + n$. The factor g may be one.

Concentration. Fix $f \sim \chi n^{it}$ and define

$$F_X(f, Y) = \sum_{Y < p \leq X} \frac{f(p)\overline{\chi(p)}p^{-it} - 1}{p}, \quad G_X(f, Y) = \sum_{Y < p \leq X} \frac{\omega(p)(f(p)\overline{\chi(p)}p^{-it} - 1)}{p},$$

where $\omega(p)$ counts roots of $1 + z^2$ modulo p . Frantzikinakis–Mountakis, [3, Propositions 2.1–2.2 and Remark 2], specialize as follows. Put $\epsilon_f(Y) = \mathbb{D}(f, \chi n^{it}; Y, \infty) + Y^{-1/2}$. Suppose B has exactly the primes at most Y as prime divisors and $\text{cond}(\chi) \mid B$. For $(b, B) = 1$,

$$\limsup_{X \rightarrow \infty} \mathbb{E}_{n \leq X} |f(Bn + b) - \chi(b)(Bn)^{it} e^{F_X(f, Y)}| \ll \epsilon_f(Y). \quad (3)$$

For $(a^2 + b^2, B) = 1$,

$$\limsup_{X \rightarrow \infty} \mathbb{E}_{m, n \leq X} \left| f((Bm + a)^2 + (Bn + b)^2) - \chi(a^2 + b^2)(B^2(m^2 + n^2))^{it} e^{G_X(f, Y)} \right| \ll \epsilon_f(Y). \quad (4)$$

Take Y sufficiently large; the distance tail is at most one in the quadratic estimate. The constants do not depend on B, a, b . For a finite family of admissible data, the maximum can be placed inside $\limsup_X$. Also $\Re F_X, \Re G_X \leq 0$ and $\epsilon_f(Y) \rightarrow 0$. These are the only concentration estimates used.

1.2 One simultaneous CRT construction

Fix integers $5 \leq r < K < Y$. Independently and uniformly select

$$q_1 \in \Phi_r = \left\{ \prod_{p \leq r} p^{e_p} : r < e_p \leq 3r/2 \right\}, \quad q_2 \in \Phi_{r,K} = \left\{ \prod_{r < p \leq K} p^{e_p} : K < e_p \leq 3K/2 \right\}.$$

Exponents are integers. Put $Q = \prod_{p \leq Y} p^{4Y}$. Choose $0 \leq v < Q$ by CRT with

$$v \equiv q_1/2 \pmod{q_1^2/2}, \tag{5}$$

$$v \equiv 1 + p^{e_p(q_2)} \pmod{p^{e_p(q_2)+1}} \quad (r < p \leq K), \tag{6}$$

$$v \not\equiv 0, \quad v^2 \not\equiv 1, -1 \pmod{p} \quad (K < p \leq Y). \tag{7}$$

The three groups of moduli have disjoint prime supports and divide Q . The last condition excludes at most five residues and $p \geq 7$. Set $M = Qm + 1$, $N = Qn + v$, using the same data for all pairs.

Exactly,

$$(2v, Q) = q_1, \quad (1 - v^2, Q) = q_2, \quad (1 + v^2, Q) = 1. \tag{8}$$

The first congruence fixes the valuation of $2v$ also at 2. For $r < p \leq K$, only $1 - v$, not $1 + v$, has valuation $e_p(q_2)$. The remaining primes divide none of the three expressions. Also

$$\frac{2v}{q_1} \equiv 1 \pmod{q_1}, \quad \frac{1 - v^2}{q_2} \equiv q_2^{-1} \pmod{q_1}, \quad 1 + v^2 \equiv 1 \pmod{q_1}. \tag{9}$$

Here $v^2 \equiv 0 \pmod{q_1}$; at 2 this follows from $2(e_2(q_1) - 1) \geq e_2(q_1)$. These are product congruences, not claims that each linear factor is separately one modulo q_1 .

1.3 Positive angular averaging and its order

Fix an integer $L \geq 2$. On $1 \leq m, n \leq X$ give weight

$$w_L(m, n) = \frac{m}{n} \mathbf{1}_{\{L \leq \log(m/n) \leq 2L\}}, \quad Z_{L,X} = \sum_{m,n \leq X} w_L(m, n).$$

Normalize by $Z_{L,X}$ and then average uniformly over q_1, q_2 . For large X this defines a finite probability average $\mathcal{A}_{L,r,K,Y,X}$. Its triples are positive: $m > n$ and $0 \leq v < Q$ imply $M > N$. The order is

$$X \rightarrow \infty, \quad Y \rightarrow \infty, \quad K \rightarrow \infty, \quad r \rightarrow \infty, \quad L \rightarrow \infty. \tag{10}$$

Every finite CRT family is fixed before X tends to infinity.

For area measure on $(0, 1)^2$, write $n = me^{-u}$ with $u > 0$. The Jacobian is me^{-u} , so integration in m gives $e^{-u} du/2$. Multiplying by $m/n = e^u$ leaves $du/2$. Hence $Z_{L,X}/X^2 \rightarrow L/2$, and the limiting weighted u distribution is uniform on $[L, 2L]$.

For fixed L the weight is bounded and Riemann integrable. Finite step approximations in u reduce cancellation to the sectors in (2). Replacing a sector in (m, n) by its version in $(Qm+1, Qn+v)$ changes $o(X^2)$ points for fixed data: after rescaling these are shrinking boundary strips. Weighted normalized concentration errors are bounded by C_L times the unweighted errors. This is harmless because L is the outermost limit. No bound uniform in L is needed.

1.4 Aperiodic characters

If f is aperiodic, the pair weights, up to the fixed factor $f(2)$ in the first two, are

$$\begin{aligned} f(M)f(N)\overline{f(M-N)}\overline{f(M+N)}, \\ f(M)f(N)\overline{f(M^2+N^2)}, \\ f(M-N)f(M+N)\overline{f(M^2+N^2)}. \end{aligned}$$

In each line an aperiodic linear factor is independent of the other linear factors. Use (2); in the first line take the quadratic weight to be one. Thus all three averages tend to zero at the inner X limit. The finite datum average and the fixed angular approximation preserve zero.

1.5 Pretentious characters: exact normalization

Fix $f \sim \chi n^{it}$. Take r so large that $\text{cond}(\chi) \mid q_1$ for every $q_1 \in \Phi_r$. The normalized linear factors are

$$M, \quad \frac{2N}{q_1} = \frac{2Q}{q_1}n + \frac{2v}{q_1}, \quad \frac{M-N}{q_2} = \frac{Q}{q_2}(m-n) + \frac{1-v}{q_2}, \quad M+N.$$

Each slope contains every prime at most Y , contains the conductor, and has no other prime factor. Each constant is coprime to its slope by (8). The product of the two Dirichlet values for $M, 2N/q_1$ is one; the product for $(M-N)/q_2, M+N$ is $\overline{\chi(q_2)}$, by (9).

Estimate (3) also controls $m-n$ and $m+n$ in the two-dimensional average, since each value has at most X representations. For $m+n$ use the interval up to $2X$; replacing F_{2X} by F_X costs $o(1)$ because $\sum_{X < p \leq 2X} 1/p \rightarrow 0$. A negative constant in the $m-n$ progression can be replaced by a positive congruent representative using a fixed shift; at most $O(X)$ boundary pairs change. The norm estimate applies with $B = Q, a = 1, b = v$, whose normalizing gcd is one.

Consequently, in normalized weighted L^1 ,

$$f(P_1(M, N)) \simeq A_f(q_1)(2Q^2mn)^{it}E_X, \quad (11)$$

$$f(P_2(M, N)) \simeq B_f(q_2)(Q^2(m^2 - n^2))^{it}E_X, \quad (12)$$

$$f(P_3(M, N)) \simeq (Q^2(m^2 + n^2))^{it}H_X, \quad (13)$$

where

$$A_f(q_1) = f(q_1)q_1^{-it}, \quad B_f(q_2) = f(q_2)\overline{\chi(q_2)}q_2^{-it}, \quad E_X = e^{2F_X(f, Y)}, \quad H_X = e^{G_X(f, Y)}.$$

The X -limsup error in each $\simeq$ is at most $C_L \epsilon_f(Y)$, uniformly over the finite datum family. Both nuisance factors have modulus at most one and are independent of *both* q_1 and q_2 . They can oscillate with X ; their convergence is neither assumed nor needed.

1.6 Independent modulus averages remove the other characters

Write $k_{ij} = \mathcal{A}_{L, r, K, Y, X} f(P_i/P_j)$. After the concentration errors vanish as $Y \rightarrow \infty$, the first two pair averages have the factor

$$\alpha_r(f) = \mathbb{E}_{q_1 \in \Phi_r} A_f(q_1),$$

while the third has the factor

$$\beta_{r, K}(f) = \mathbb{E}_{q_2 \in \Phi_{r, K}} B_f(q_2).$$

Every remaining factor has modulus at most one. In pair 12 the nuisance factor is $|E_X|^2$; in pairs 13 and 23 it is $E_X \overline{H_X}$. The powers of Q cancel, and the angular ratios depend only on m/n , so these factors are independent of the auxiliary moduli.

Unless $f(n) = n^{it}$ exactly, there is a fixed prime p with $f(p)p^{-it} \neq 1$. Its independent exponent interval in q_1 has length tending to infinity as $r \rightarrow \infty$. The geometric-sum bound therefore gives $\alpha_r(f) \rightarrow 0$. Thus both k_{12} and k_{13} vanish unless f is exactly Archimedean.

If f is not eventually χn^{it} on primes, then for every fixed r there is a mismatch prime $p > r$. Once $K \geq p$, its exponent interval in q_2 grows with K , giving $\beta_{r,K}(f) \rightarrow 0$. This removes pair 23 outside $\mathcal{E}$. For $f \in \mathcal{E}$, choose an eventual model and take r beyond its exceptional primes and conductor. Now P_2, P_3 avoid all primes at most r and are both one modulo the conductor. Thus, exactly,

$$f(P_2/P_3) = (P_2/P_3)^{it}. \quad (14)$$

Every survivor is handled without a residual concentration phase. The cutoff can depend on f ; no uniform conductor bound is needed.

1.7 The final angular limit

For $f(n) = n^{it}$ the first two limiting angular kernels are

$$\frac{1}{L} \int_L^{2L} \exp\left(it \log \frac{2e^u}{e^{2u} - 1}\right) du, \quad \frac{1}{L} \int_L^{2L} \exp\left(it \log \frac{2e^u}{e^{2u} + 1}\right) du.$$

The logarithmic phases are $\log 2 - u + O(e^{-2u})$. For fixed $t \neq 0$ their moduli are at most $2/(|t|L) + O(|t|e^{-2L})$, hence tend to zero; at $t = 0$ they are one. The fixed CRT shifts disappear in the inner scaled-box limit. This gives the first two targets in (1).

For (14), the kernel is

$$\frac{1}{L} \int_L^{2L} \exp(it \log(\tanh u)) du \rightarrow 1,$$

since $\log(\tanh u) = O(e^{-2u})$. This gives the third target. The kernels are bounded by one. Thus continuous Archimedean spectral mass is handled as well as atoms; it is not approximated by finitely many eigencharacters.

1.8 All three spectral limits use the same finite averages

For the finite kernels define

$$e_{L,r,K,Y,X}(f) = |k_{12}(f) - \mathbf{1}_{\{1\}}(f)| + |k_{13}(f) - \mathbf{1}_{\{1\}}(f)| + |k_{23}(f) - \mathbf{1}_{\mathcal{E}}(f)|.$$

The preceding estimates prove, for every f ,

$$\limsup_L \limsup_r \limsup_K \limsup_Y \limsup_X e_{L,r,K,Y,X}(f) = 0, \quad 0 \leq e \leq 6. \quad (15)$$

All parameters are integers and every finite kernel is continuous. The set $\mathcal{E}$ is Borel independently of this calculation. Fix a Dirichlet character, a finite exceptional prime set containing its conductor primes, and a compact interval for t . The corresponding characters form a compact set: they are the continuous image of that interval times the finitely many free unit circles at exceptional primes. Taking the countable union over these choices gives $\mathcal{E}$.

For any finite Borel measure ν on $\mathcal{M}$, bounded reverse Fatou at each of the five limits yields

$$\begin{aligned} & \limsup_L \limsup_r \limsup_K \limsup_Y \limsup_X \int e_{L,r,K,Y,X}(f) d\nu(f) \\ & \leq \int \limsup_L \limsup_r \limsup_K \limsup_Y \limsup_X e_{L,r,K,Y,X}(f) d\nu(f) = 0. \end{aligned}$$

For instance $\limsup_X \int e \leq \int \limsup_X e$; the new integrand remains bounded by six, permitting repetition. The restrictions $r < K < Y$ remove only finite initial segments. Therefore the *same* finite positive averages approximate all three integrated limits simultaneously. No unbounded limit exchange or uniform rate on $\mathcal{M}$ has been invoked.

1.9 An integer coloring supplies the required spectrum

Use the multiplicative boxes

$$\Psi_j = \left\{ \prod_{p \leq j} p^{e_p} : 0 \leq e_p \leq j \right\} \subset \mathbb{N}.$$

For fixed $a \geq 1$, $|a\Psi_j \triangle \Psi_j|/|\Psi_j| \rightarrow 0$: after all prime factors of a appear, only a bounded number of exponent boundary layers is lost. A weak-star cluster point of the box averaging functionals is a positive normalized mean $\mathfrak{m}$ on $\ell^\infty(\mathbb{N})$ with $\mathfrak{m}_n h(an) = \mathfrak{m}_n h(n)$.

For a sign coloring c put

$$\rho(a/b) = \mathfrak{m}_n c(an) c(bn).$$

This is well-defined. If $a/b = a'/b'$, dilating the first expression by b' and the second by b makes the products identical. Also $\rho(1) = 1$, and ρ is positive definite: clear the common denominators of finitely many rationals r_i to obtain integers A_i with $r_i/r_j = A_i/A_j$. Then

$$\sum_{i,j} z_i \bar{z}_j \rho(r_i/r_j) = \mathfrak{m}_n \left| \sum_i z_i c(A_i n) \right|^2 \geq 0.$$

Bochner–Herglotz on the discrete abelian group $\mathbb{Q}_{>0}$ supplies a probability measure ν on $\mathcal{M}$ with $\rho(q) = \int f(q) d\nu(f)$.

Suppose there were no monochromatic Pythagorean triple. For every positive integer shape $a^2 + b^2 = c_0^2$ and every n , two of the three sign pairs are opposite and one is equal. Hence

$$c(an)c(bn) + c(an)c(c_0 n) + c(bn)c(c_0 n) = -1.$$

Applying the mean gives $\rho(a/b) + \rho(a/c_0) + \rho(b/c_0) = -1$. Average over the common Euclidean triples above. By the proved spectral integration, the left side tends to

$$2\nu(\{1\}) + \nu(\mathcal{E}) \geq 0,$$

whereas it equals -1 at every finite stage. This proves Theorem 1.1.

In fact the monochromatic indicator of three signs is $(1 + s_1 s_2 + s_1 s_3 + s_2 s_3)/4$. Its common averaged mean is $(1 + 2\nu(\{1\}) + \nu(\mathcal{E}))/4 \geq 1/4$. Some fixed positive integer Euclidean shape therefore has positive mean of its monochromatic integer dilations, and hence at least one such dilation. This also explains the rational-to-integer interface explicitly. There is no bound here for the smallest triple.

For three or more colors the binary identity fails. Pairwise independent uniform three-color labels can be supported on the eighteen patterns with exactly two equal coordinates, with no monochromatic triple. Thus this proof does not resolve arbitrary finite colorings.

2 The Gaussian cancellation input: a direct repair

The Gaussian cancellation needed in the common-pair argument follows from the older Frantzikinakis–Host theorem. It does not require the disputed four-point recurrence statement. We give the reduction, including affine grids, and record a defect in the newer written proof. The deep input retained here is the Gowers uniformity theorem for aperiodic multiplicative functions, together with its linear-form consequence: Frantzikinakis–Host, *Higher order Fourier analysis of multiplicative functions and applications*, Theorem 2.5 and Lemma 9.6, <https://arxiv.org/pdf/1403.0945>. Their Theorem 9.7 also directly contains the required Gaussian case. This section is not a new proof of that deep uniformity theorem.

Proposition 2.1 (Gaussian cancellation on fixed affine grids). *Let $L_1, \dots, L_s$ be nonzero integral linear forms in two variables, with L_1 independent of every L_j , $j > 1$. Let f_1 be an aperiodic multiplicative function bounded by one, extended evenly to $\mathbb{Z}$, and let $f_j : \mathbb{Z} \rightarrow \mathbb{C}$, $j > 1$, be fixed functions bounded by one. Let g be completely multiplicative and bounded by one. For a fixed bounded periodic function $\omega : \mathbb{Z}^2 \rightarrow \mathbb{C}$ and arbitrary convex $K_N \subset [-cN, cN]^2$, where c is fixed, one has*

$$\frac{1}{N^2} \sum_{(x,y) \in \mathbb{Z}^2} \mathbf{1}_{K_N}(x,y) \omega(x,y) g(x^2 + y^2) \prod_{j=1}^s f_j(L_j(x,y)) \longrightarrow 0.$$

The origin may be omitted. In particular, this holds on every fixed grid $(x,y) = (Qm + a, Qn + b)$ with a convex sector cutoff.

Periodic factors. Expand ω in its finite additive Fourier series. Choose integral forms A, B independent of each other and each independent of L_1 . For each Fourier frequency θ there are real α, β with $\theta(x,y) = \alpha A(x,y) + \beta B(x,y)$. Append $e(\alpha A)$ and $e(\beta B)$ as two bounded nondistinguished factors. Thus it suffices to consider $\omega = 1$. The distinguished function remains even and multiplicative. This observation already reduces the proposition to the convex-set version of Frantzikinakis–Host Theorem 9.7. The following argument checks the Gaussian reduction directly.

Proof. Write $z = x + iy$, $L_j(z) = \text{Im}(\zeta_j z)$, where $\zeta_j = b_j + ia_j$ if $L_j(x,y) = a_j x + b_j y$. After a fixed rescaling of N , let $B_N = \{x + iy : |x|, |y| \leq N\}$ and put

$$a_N(z) = \mathbf{1}_{K_N}(z) \prod_j f_j(L_j(z)), \quad G(z) = g(|z|^2) \quad (z \neq 0), \quad G(0) = 0.$$

Extend a_N by zero outside B_N . All zero arguments of the linear forms contribute only $O(N)$ points if their assigned values are changed. The function G is completely multiplicative.

For every rational prime $p \equiv 1 \pmod{4}$, choose one Gaussian prime π of norm p , omitting the finitely many primes dividing any $|\zeta_j|^2$. The sum of reciprocal norms diverges. Fix a finite collection Π of these primes and set

$$H = \sum_{\pi \in \Pi} \frac{1}{|\pi|^2}, \quad w(z) = \sum_{\pi \in \Pi} \mathbf{1}_{\{\pi|z\}}.$$

Lattice counts and the Chinese remainder theorem give, for this fixed collection,

$$\mathbb{E}_{B_N} w = H + o(1), \quad \mathbb{E}_{B_N} (w - H)^2 = \sum_{\pi \in \Pi} \left(\frac{1}{p} - \frac{1}{p^2} \right) + o(1) \leq H + o(1).$$

Put $S_N = \sum_z a_N(z) G(z)$ and $T_N = \sum_z a_N(z) G(z) w(z)$. Then

$$|HS_N - T_N| \leq |B_N| \sqrt{H + o(1)}.$$

Complete multiplicativity gives the exact reindexing

$$T_N = \sum_u G(u) \sum_{\pi \in \Pi} G(\pi) a_N(\pi u).$$

Every relevant u lies in B_{2N} . Cauchy–Schwarz bounds $|T_N|^2$ by $|B_{2N}|$ times the sum of the corresponding diagonal and off-diagonal correlations. The diagonal is at most

$$\sum_{\pi \in \Pi} \sum_u |a_N(\pi u)|^2 \leq |B_N|H + o(N^2).$$

We claim that each off-diagonal correlation is $o(N^2)$.

Fix distinct $\pi, \rho \in \Pi$. Its forms are $L_j(\pi u), L_j(\rho u)$, and its cutoff is the convex set $\pi^{-1}K_N \cap \rho^{-1}K_N$. The distinguished form $L_1(\pi u)$ is independent of all other forms with multiplier π . Proportionality with $L_j(\rho u)$ would yield

$$k\zeta_1\pi = l\zeta_j\rho \quad (k, l \in \mathbb{Z} \setminus \{0\}).$$

At the two distinct Gaussian primes $\pi, \bar{\pi}$ above p , the valuations of a rational integer agree. The displayed equality would therefore imply both $v_p(k) + 1 = v_p(l)$ and $v_p(k) = v_p(l)$, a contradiction. All other factors have zero valuation at these two primes by the choice of the finite excluded set and the distinct rational norms.

The distinguished f_1 has vanishing Gowers norm of every fixed order by the older uniformity theorem. The linear-form estimate, with the independence just proved, now makes the off-diagonal correlation $o(N^2)$. It applies to arbitrary convex cutoffs and bounded other factors. Signed boxes can be split into four quadrants and reflected; the axes contribute $O(N)$ points.

Consequently, since $|B_{2N}|/|B_N| \rightarrow 4$,

$$\limsup_{N \rightarrow \infty} \frac{|S_N|}{|B_N|} \leq \frac{3}{\sqrt{H}}.$$

Finally enlarge the finite collection so that $H \rightarrow \infty$. All constants and errors may depend on the finite collection, which was fixed before $N \rightarrow \infty$. This proves the proposition, allowing the cutoff to depend on N . $\square$

Convex-cutoff uniformity. The elementary localization behind the linear-form estimate is uniform over convex subsets of a bounded square: a mesh of side ϵN has $O(\epsilon N^2 + N)$ lattice points in its boundary cells. For fixed ϵ only finitely many cells are involved. On each cell, the cyclic-group generalized von Neumann estimate and the even extension of f_1 give convergence to zero. First let $N \rightarrow \infty$ and then $\epsilon \rightarrow 0$. Equivalently, smooth the cutoff at a fixed relative scale and truncate its Fourier expansion before taking the large-box limit. This explains why a changing sequence of convex sets is permitted.

Affine grids and the actual three applications. For fixed Q, a, b , the change $x = Qm + a$, $y = Qn + b$ gives the rectangle

$$[Q + a, QN + a] \times [Q + b, QN + b]$$

with periodic weight $\mathbf{1}_{\{x \equiv a, y \equiv b \pmod{Q}\}}$. A sector in (x, y) intersects this rectangle in a convex set. A sector defined originally by $u \leq m/n \leq v$ also gives affine halfplanes $u(y - b) \leq x - a \leq v(y - b)$

and hence a convex set, on the positive- n rectangle. The ambient scale is $O_Q(N)$. No primitivity condition on the residues is needed.

The common-pair proof uses $(x, y, x - y, x + y)$ with $g = 1$, and (x, y) or $(x - y, x + y)$ with a Gaussian norm weight. In every case the distinguished aperiodic factor is independent of the other forms; conjugation preserves aperiodicity. Every CRT family is finite and fixed at the inner limit. A bounded angular weight is approximated by finitely many sectors.

The defect being avoided. In the generalized FKM v3 Section 3.4, <https://arxiv.org/html/2407.08360v3>, the sets R_N are positive-quadrant rectangles, but the claimed containment $z^{-1}R_N \subset R_{CN/\sqrt{N(z)}}$ is false: $(1 + i)(2 - i) = 3 + i \in R_3$, whereas $2 - i$ lies in no R_M . This is a defect in the written reduction, not a counterexample to Gaussian cancellation. The proof above uses symmetric domains and an explicit finite criterion for size-dependent cutoffs. It certifies the needed Gaussian case only, not every statement of the generalized paper or the separate four-point recurrence claim.

References

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